

Lesson:

Visualization of DOM & It's Working



Topics Covered

1. What is DOM?
2. Visualization on DOM
3. step-by-step guide to visualizing the DOM using the Chrome browser
4. Working of DOM

What is DOM?

The Document Object Model (DOM) is a programming interface for web documents. It represents the web page as a tree-like structure of objects, where each object corresponds to an element or attribute in the web page. The DOM provides a way for JavaScript to interact with and manipulate the elements and content of a web page.

When a web page is loaded in a browser, the browser parses the HTML and creates the DOM tree, which is a live representation of the page's structure. This tree-like structure allows JavaScript to interact with and modify the content, structure, and styles of the web page, making it more interactive and dynamic.

Visualization on DOM

By visualizing the DOM, we can see the hierarchical structure of the HTML document. Each element is represented as a node, and elements that are nested inside other elements become child nodes. Understanding the DOM tree helps web developers access and manipulate specific elements using JavaScript, allowing them to build dynamic and interactive web pages. The hierarchical nature of the DOM is fundamental to creating responsive web applications where elements can be modified, added, or removed dynamically based on user interactions and other events.

In most web browsers, you can open the developer tools by pressing the F12 key or right-clicking on the web page and selecting "Inspect" or "Inspect Element". Once the developer tools are open, you can navigate to the "Elements" or "DOM" tab to view the DOM tree.

The DOM tree is displayed as a hierarchical structure of nodes, with each node representing an HTML or XML element, attribute, or text content. You can expand or collapse nodes to view their children or siblings, and you can hover over a node to highlight the corresponding element on the web page.

You can also use the developer tools to modify the DOM in real-time. For example, you can add, delete, or modify nodes, attributes, and text content by clicking on the relevant element and using the editor that appears.

In addition, the developer tools allow you to view the CSS styles that are applied to each element, as well as to modify the styles and see the effects on the web page in real-time.

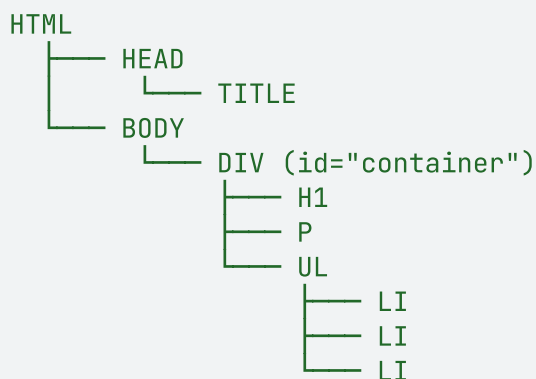
Overall, visualizing the DOM in JavaScript involves using the web browser's developer tools to view and manipulate the tree-like structure of objects that represent the HTML or XML document in the browser. This can be a powerful tool for debugging and understanding the structure and content of a web page.

Example of DOM:

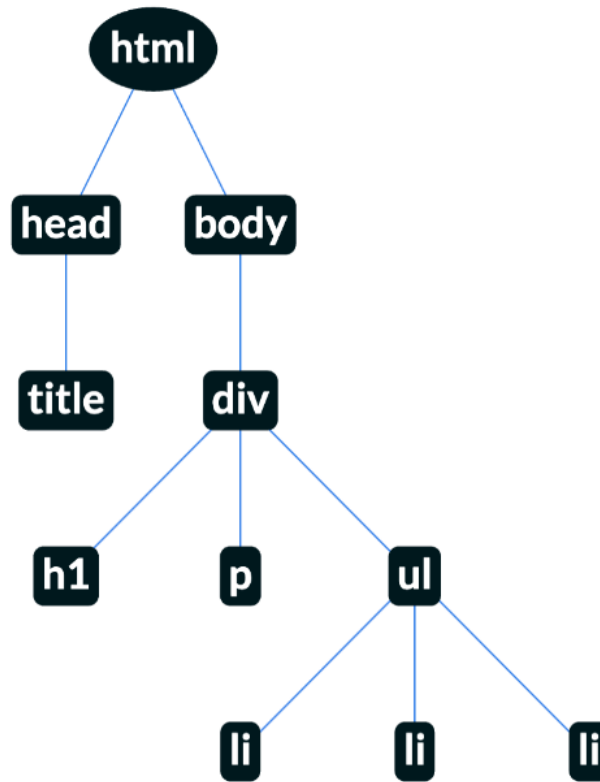
This is our HTML code.

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Visualization</title>
</head>
<body>
  <div id="container">
    <h1>Hello, World!</h1>
    <p>This is a sample paragraph.</p>
    <ul>
      <li>Item 1</li>
      <li>Item 2</li>
      <li>Item 3</li>
    </ul>
  </div>
</body>
</html>
```

And this will be our DOM representation:



DOM Tree



step-by-step guide to visualizing the DOM using the Chrome browser

To visualize the Document Object Model (DOM) in JavaScript, you can use the developer tools in your web browser. Here's a step-by-step guide to visualizing the DOM using the Chrome browser:

- Open the Chrome browser and navigate to the web page you want to inspect.
- Press the F12 key to open the developer tools.
- In the developer tools, select the "Elements" tab to view the DOM tree.
- You can expand or collapse nodes to view their children or siblings.
- To select an element in the DOM, click on it in the "Elements" tab or on the web page itself. The corresponding element in the DOM tree will be highlighted.
- You can also modify the DOM in real-time using the developer tools. To add, delete, or modify an element or attribute, right-click on the relevant node and select "Edit as HTML" or "Edit attribute", then make your changes in the editor that appears.
- To view or modify the CSS styles that are applied to each element, select the "Styles" tab in the developer tools. You can view and modify the styles and see the effects on the web page in real time.

Other web browsers, such as Firefox and Safari, have similar developer tools for visualizing and manipulating the DOM. By visualizing the DOM in JavaScript, you can better understand the structure and content of a web page, and use this knowledge to write more effective JavaScript code.

Working of DOM

- The Document Object Model (DOM) is a programming interface for web documents that is used by JavaScript to interact with and manipulate the elements and content of a web page. Here's a detailed explanation of how the DOM works in JavaScript:
- When a web page is loaded in a web browser, the browser creates a DOM tree that represents the structure and content of the web page. The DOM tree is a hierarchical data structure that is composed of nodes, where each node corresponds to an element, attribute, or text content in the web page. The root of the tree is the "document" object, which represents the entire web page.
- Each node in the DOM tree is an object that has properties and methods that can be accessed and manipulated by JavaScript. For example, the "document" object has properties and methods that allow JavaScript to access and manipulate the elements and content of the web page.
- To interact with the DOM in JavaScript, you can use various methods and properties that are provided by the DOM API. For example, to select an element in the DOM, you can use the "getElementById()", "querySelector()", or "getElementsByTagName()" methods. These methods return a reference to the selected element as a DOM node, which can then be used to access or modify the properties of the element.
- Once you have a reference to a DOM node, you can use its properties and methods to access or modify its content, style, or other attributes. For example, you can use the "innerHTML" property to get or set the HTML content of an element, the "style" property to access or modify the CSS style of an element, or the "setAttribute()" method to set the value of an attribute on an element.
- When you modify the properties or content of a DOM node in JavaScript, the corresponding element on the web page is updated in real-time, without the need to reload the web page. This enables dynamic web development, where web pages can respond to user interactions, events, or data changes in real time, without requiring a server-side refresh.
- Overall, the DOM provides a powerful and flexible programming interface for web documents that is used by JavaScript to interact with and manipulate the elements and content of a web page and is a key component of modern web development.